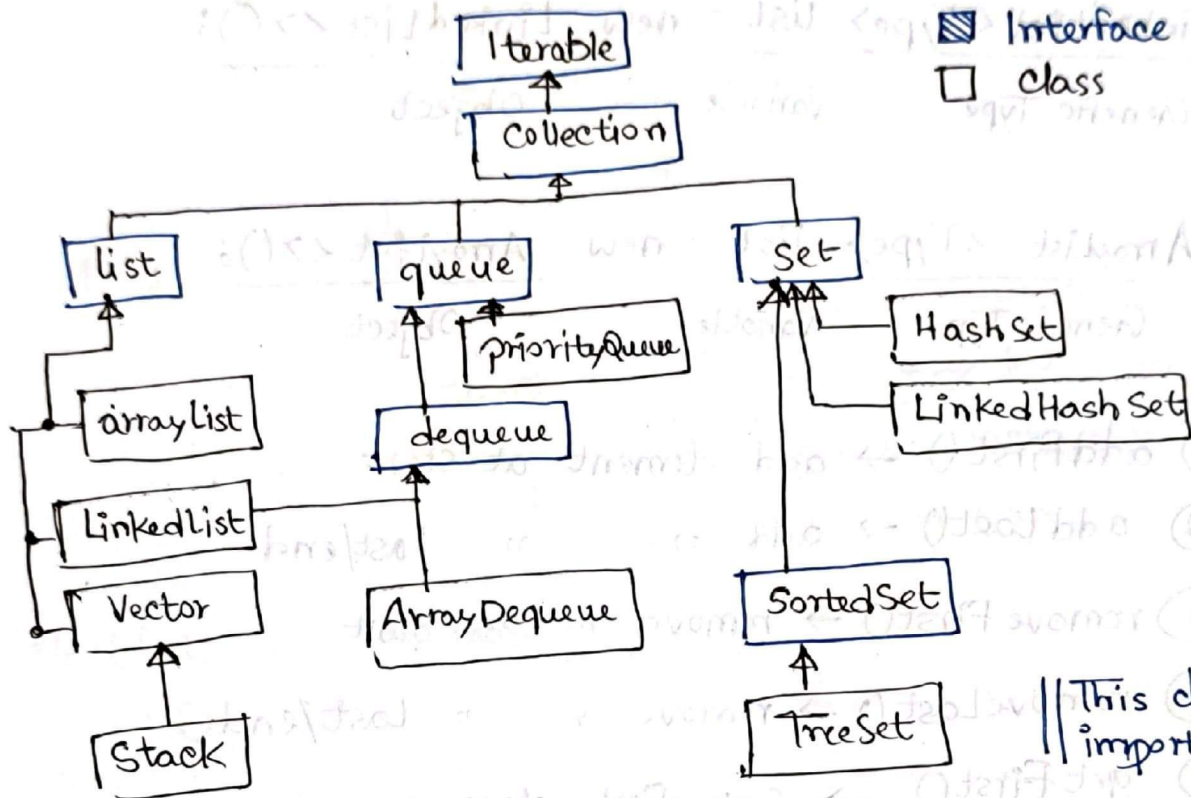


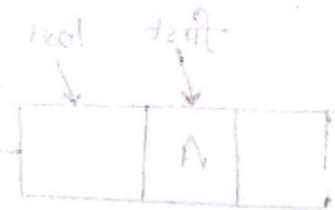
String handling → sprint (शक्ति)

Java Collection Framework



ArrayList vs. LinkedList

- | | |
|--|--|
| <ul style="list-style-type: none"> ↳ Dynamic Allocation Possible ↳ Duplicate elements possible | <ul style="list-style-type: none"> ↳ Static Can't add new items and expand its size |
|--|--|



Linked List

Singly, Doubly, Circular

↓
Java

LinkedList efficiency > Array efficiency

LinkedList Methods

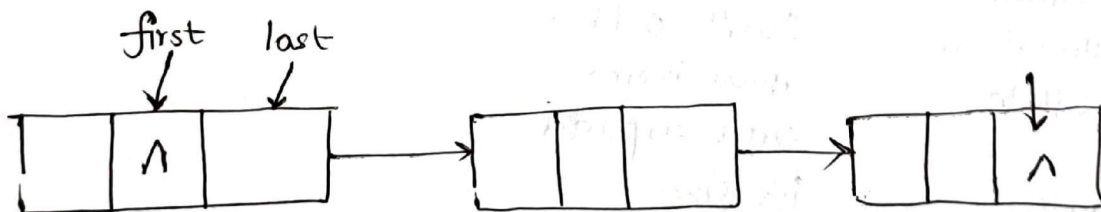
java.util.*

General Syntax :

LinkedList<Type> list = new LinkedList<>();
Generic Type Variable Object

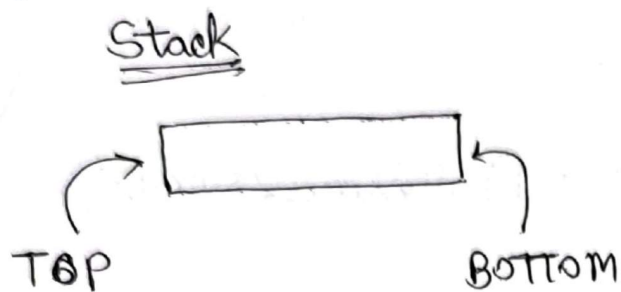
ArrayList<Type> list = new ArrayList<>();
Generic Type Variable Object

- ① addFirst() → add element at start
- ② addLast() → add „ „ Last/end
- ③ removeFirst() → remove „ „ start
- ④ removeLast() → remove „ „ Last/end
- ⑤ getFirst() → Get first element
- ⑥ getLast() → Get last element

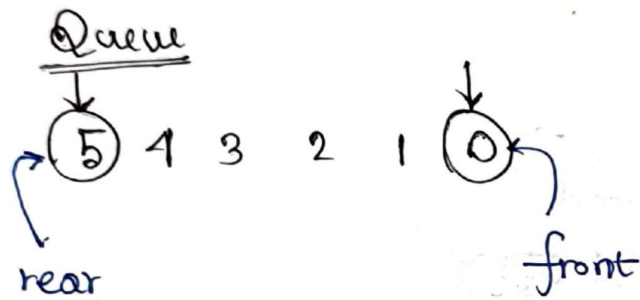
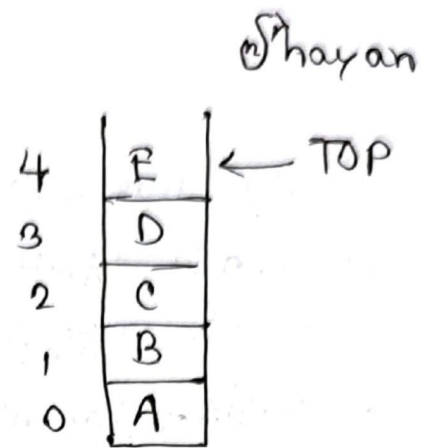


- ⑦ add()
- ⑧ clear()
- ⑨ get()/element()
- ⑩ indexOf()
- ⑪ lastIndexOf()

- ⑫ peek()
- ⑬ peekFirst()
- ⑭ peekLast()
- ⑮ poll → retrieve & remove
- ⑯ push()
- ⑰ pop()



Stack Empty/No element = -1



Priority Queue

Priority is set using
Alphabetically by default
But can be customized
with custom class.

Set (int index, Element e)
to Array()
size()

```
Iterator <String> itr = al.iterator();
while (itr.hasNext()) {
    Sout (itr.next());
}
```

Stack → LIFO
Queue → FIFO

Queue Method

- ① add
- ② offer
- ③ remove / poll
- ④ element
- ⑤ peek

```
PriorityQueue <String> queue = new PriorityQueue <String> ();
queue.add ("Amit");
queue.add ("Vijay");
queue.add ("Karan");
queue.add ("Jai");
```

PriorityQueue ব্যাখ্যা

- ↳ order is not maintained
- ↳ even if order is add করা হয়েছে
হুটাত্ত আসা না।
- ↳ use " queue.isEmpty() "

Stack

- ① Empty
- ② push ← "top"
- ③ pop ← "top"
- ④ peek
- ⑤ search

↑ search specific object
and return its position

```
Stack <Integer> stk = new Stack <> ();
boolean result = stk.empty();
```

<https://tpointtech.com>

Iterator

- ① hasNext()
- ② next()
- ③ remove()
- ④ forEachRemaining()

Scenario-based
what data structure to chose?